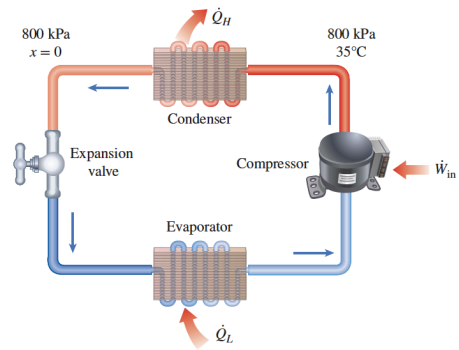


Chapter 6 Workshop

1. Refrigerant-134a enters the condenser of a residential heat pump at 800 kPa and 35°C at a rate of 0.018 kg/s and leaves at 800 kPa as a saturated liquid. If the compressor consumes 1.2 kW of power, determine (a) the COP of the heat pump and (b) the rate of heat absorption from the outside air.



R-134a

$$P_1 = 800 \text{ kPa}$$

$$T_1 = 35^\circ\text{C}$$

$$\dot{m} = 0.018 \text{ kg/s}$$

$$P_2 = 800 \text{ kPa}$$

Sat. liq

For h_{in} :

$$P_1 = 800 \text{ kPa}$$

$$T_1 = 35^\circ\text{C}$$

A-12

$$T_1 > T_{sat} \rightarrow \text{superheated}$$

A-13

interpolation

$$h_1 = (267.34) + (35 - 31.31) \left(\frac{276.46 - 267.34}{40 - 31.31} \right) = 271.21$$

For h_{out} :

$$P_2 = 800 \text{ kPa}$$

Sat. liq

A-12

$$h_f = 95.48 \frac{\text{kJ}}{\text{kg}}$$

$$\dot{Q}_H = \dot{m} (h_{in} - h_{out}) = (0.018) (271.21 - 95.48) \times$$

$$\Rightarrow \dot{Q}_H = 3.16 \text{ kW}$$

$$\text{COP}_{HP} = \frac{\dot{Q}_H}{\dot{W}_{in}} = \frac{3.16}{1.2} = 2.64$$

2. Heat is transferred to a heat engine from a furnace at a rate of 80 MW. If the rate of waste heat rejection to a nearby river is 50 MW, determine the net power output and the thermal efficiency for this heat engine.

SOLUTION The rates of heat transfer to and from a heat engine are given. The net power output and the thermal efficiency are to be determined.

Assumptions Heat losses through the pipes and other components are negligible.

Analysis A schematic of the heat engine is given in Fig. 6–16. The furnace serves as the high-temperature reservoir for this heat engine and the river as the low-temperature reservoir. The given quantities can be expressed as

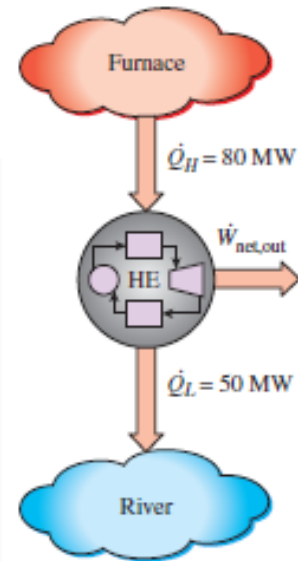
$$\dot{Q}_H = 80 \text{ MW} \quad \text{and} \quad \dot{Q}_L = 50 \text{ MW}$$

The net power output of this heat engine is

$$\dot{W}_{\text{net,out}} = \dot{Q}_H - \dot{Q}_L = (80 - 50) \text{ MW} = 30 \text{ MW}$$

Then the thermal efficiency is easily determined to be

$$\eta_{\text{th}} = \frac{\dot{W}_{\text{net,out}}}{\dot{Q}_H} = \frac{30 \text{ MW}}{80 \text{ MW}} = 0.375 \text{ (or 37.5\%)}$$



3. During an experiment conducted in a room at 25°C, a laboratory assistant measures that a refrigerator that draws 2 kW of power has removed 30,000 kJ of heat from the refrigerated space, which is maintained at -30°C. The running time of the refrigerator during the experiment was 20 min. By comparing actual and theoretical maximum COPs, determine if these measurements are reasonable.

